Whole Genome Sequencing: Rapid Trio

Proband Report

BAYLOR GENETICS

DEMOGRAPHIC INFORMATION

PATIENT

NAME:	[redacted]
DATE OF BIRTH:	[redacted]
SEX:	[redacted]
MEDICAL RECORD #:	EP03618349
ACCESSION #:	
LAB NUMBER:	[redacted]
FAMILY NUMBER:	[redacted]

TEST INFORMATION

TEST NAME:	Rapid Trio WGS
TEST CODE:	1822
SAMPLE TYPE:	[redacted]
DATE COLLECTED:	[redacted]
DATE RECEIVED:	[redacted]
DATE REPORTED:	09/02/2025

RECIPIENT

PHYSICIAN NAME:	ELYSSA LEWIS
FACILITY:	NCGC
LOCATION:	LOUISVILLE KY 40202
PHONE:	502-588-0850
FAX:	502-588-0861

CLINICAL INDICATION

Based on the submitted clinical information, the patient has scoliosis, rhizomelia, vertebral segmentation defect, duodenal atresia, intestinal malrotation, absent gallbladder, polyhydramnios, short long bone, thin ribs, epicanthus, prominent forehead, deeply set eye, anteverted nares, low-set ears, cupped ear, pointed chin, renal insufficiency, hydronephrosis, patent foramen ovale, patent ductus arteriosus, tricuspid regurgitation and right ventricular dilatation. Family history is significant for a paternal half-aunt with a congenital heart defect, a paternal aunt with cardiac symptoms, and a paternal cousin with congenital heart disease, a paternal grandfather with bladder cancer and heart attack, and a paternal grandmother with mental health concerns.

We have also received a sample from the father (DNA#3455879) and the mother (DNA#3455862) of this individual.

RESULTS

INDETERMINATE FINDINGS

DISEASE	INHERITANCE PATTERN	GENE / VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
SLC26A2-Related Disorders	Autosomal Recessive	SLC26A2: c.835C>T, p.R279W	Sequence Variant	Heterozygous	Father	Pathogenic

RESULTS SUMMARY

No significant findings were identified. A heterozygous pathogenic variant in the SLC26A2 gene was detected, associated with SLC26A2-related disorders; no second variant was identified.

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Houston, TX 77021-2024
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F 1.800.434.9850
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ADDITIONAL FINDINGS

ACMG recommended secondary findings (v3.2; PMID 37347242)

None Detected.

Other Incidental Findings

Client has not selected to receive these findings.

Research Findings

Client has not selected to receive these findings.

RECOMMENDATIONS

• It is recommended that correlation of these findings with the clinical phenotype be performed.
• Genetic counseling is recommended.
• For sequence variants, testing is available for this individual's at risk relatives under test code 1580. Visit https://www.baylorgenetics.com/known-familial-mutations-policy/ for further information. Please consult with the lab for available test codes for other types of variants (CNV, STR, mito).
• Contact laboratory for requests regarding reanalysis and raw data.

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INTERPRETATION - PHENOTYPE RELATED FINDINGS

DISEASE	INHERITANCE PATTERN	GENE / VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
SLC26A2-Related Disorders	Autosomal Recessive	SLC26A2: c.835C>T, p.R279W	Sequence Variant	Heterozygous	Father	Pathogenic

GENE DESCRIPTION

The SLC26A2 gene encodes a sulfate transporter which is a transmembrane glycoprotein that plays a critical role in sulfation of proteoglycans and matrix organization in cartilage. It is also known as diastrophic dysplasia sulfate transporter and is implicated in the pathogenesis of several human chondrodysplasias.

DISEASE DESCRIPTION

Defects in the SLC26A2 gene may cause a spectrum of autosomal recessive skeletal dysplasias, including achondrogenesis Ib (OMIM: 600972), which is characterized by short limbs with short fingers and toes, hypoplasia of the thorax, narrow chest, fetal hydrops and respiratory insufficiency. Defects in SLC26A2 may also cause autosomal recessive atelosteogenesis, type II and De la Chapelle dysplasia (OMIM: 256050), characterized by rhizomelic limb shortening with normal-sized skull, hitchhiker thumbs, narrow chest, cleft palate, and distinctive facial features (midface retrusion, depressed nasal bridge, epicanthus, micrognathia). Other typical findings are scoliosis, clubfoot, pulmonary hypoplasia and respiratory insufficiency. In addition, defects in the SLC26A2 gene may cause autosomal recessive epiphyseal dysplasia, multiple, 4 (OMIM: 226900) which is characterized by flattened femoral epiphyses, joint pain, hip dysplasia, clubfeet and scoliosis. Defects in SLC26A2 may also cause autosomal recessive diastrophic dysplasia (OMIM:222600), which is characterized by osteochondrodysplasia with clinical features including short limbs, short stature, cystic lesions of the pinnae, cleft palate, scoliosis, hitchhiker thumb, clubfeet and laryngotracheal stenosis.

VARIANT DETAILS - SLC26A2: c.835C>T, p.R279W

- (GRCh38)chr5:149980428C>T - NM_000112.4
- The c.835C>T (p.R279W) variant in the SLC26A2 gene has been described as (likely) pathogenic in ClinVar (ID: 4089). This alteration is the most common variant in the European, Non-Finnish population (PMID: 10465113, 11241838). This variant has been previously reported in a homozygous state in individuals with multiple epiphyseal dysplasia (PMID: 12525546) or in a heterozygous state in individuals with atelosteogenesis type II (PMID: 8571951) and diastrophic dysplasia (PMID: 11241838).
- This variant has been observed with a frequency of 0.110% in the gnomAD v4 dataset with 2 homozygote.
- A function study demonstrated that this variant causes reduced sulfate transport activity (PMID: 15294877).
- This missense variant occurs at a residue with high evolutionary conservation and this effect is predicted to be deleterious (REVEL = 0.917, CADD = 28.6).

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QUALITY METRICS

Percentage depth of coverage at 20x: **99.76%**
Mean depth of coverage: **63.91x**

METHODOLOGY

Genomic DNA was fragmented and indexed to prepare individual libraries. The libraries were purified and pooled for sequence analysis on the Illumina NovaSeq platform for paired end reads. Internal quality control measures are performed to ensure sample integrity.

Data analysis and interpretation are performed by the Baylor Genetics analytics pipeline. Data are aligned to the human reference genome build GRCh38 using the Illumina Dragen BioIT Platform. Variant calling is performed using the Illumina Dragen haplotype-based variant calling system and Illumina Dragen genome wide depth based CNV (Copy Number Variants) caller with custom modifications from Baylor Genetics. Short tandem repeat calling is performed using the Illumina Expansion Hunter tool.

The automated genetic interpretation platform from Emedgene is implemented for variant annotation and interpretation with proprietary algorithms and open-source data sets such as gnomAD, EVS (Exome Variant Server), ClinVar, and professional resources such as HGMD (Human Gene Mutation Database) Pro. The variants were interpreted according to ACMG (American College of Medical Genetics) guidelines (reference 1) and patient phenotypes.

Variants are confirmed by Sanger sequencing, specific PCR, Multiplex Ligation-dependent Probe Amplification (MLPA), or another methodology, if sequencing or copy number variants detected by NGS do not meet internal quality standards or are within highly homologous regions.

Synonymous variants, noncoding variants not affecting splicing sites, and common benign variants are excluded from interpretation unless previously reported in the literature as possibly pathogenic variants. This test may not provide detection of certain variants or portions of certain genes due to insufficient sequencing depth, local sequence characteristics, high/low genomic complexity, or the presence of closely related pseudogenes. Small deletions or insertions larger than 25bp, low-level mosaic variants, structural variants such as inversions, and/or balanced translocations may not be detected with this technology.

Variant types being analyzed include sequencing variants and CNV in nuclear genes and mitochondrial DNA (mtDNA), and short tandem repeats (STR). Uniparental disomy (UPD) analysis is done only in a trio setting. Regions of homozygosity (ROH) are reported when greater than 5 Mb. Sensitivity of detection is decreased for mtDNA sequencing variants and CNVs at heteroplasmy levels lower than 5% and 10%, respectively. Pathogenic events of short tandem repeats can be detected within the genomic regions of following genes: AFF2, AR, ATN1, ATXN1, ATXN2, ATXN3, ATXN7, ATXN10, ATNX8OS, C9orf72, CACNA1A, CNBP, CSTB, DIP2B, DMPK, FGF14, FMR1, FXN, GLS, HTT, JPH3, NOP56, NOTCH2NLC, PABPN1, PHOX2B, PPP2R2B, RFC1, TBP, and TCF4. Sensitivity of detection is reduced for borderline or incomplete penetrance alleles.

The proband report contains results of genes related to the patient's clinical phenotype. Variants relevant to the phenotype are reported as positive (pathogenic/likely pathogenic) or variants of uncertain significance (VUS). Benign and likely benign variants are not reported. This interpretation is based on the clinical information provided by the referring institution and our current understanding of the gene and variants at the time of reporting. Additional methodologies may be used to aid in variant interpretation. Results in the report should be carefully correlated to the individual's clinical findings by the referring physician.

If elected, secondary findings including pathogenic and likely pathogenic variants in genes included in the ACMG secondary findings recommendations, or other findings determined to be severe and medically significant (incidental findings), will be reported. The ACMG recommendations (ACMG SF v3.2 (reference 2)) include the following genes: ACTA2, ACTC1, ACVRL1, APC, APOB, ATP7B, BAG3, BMPR1A, BRCA1, BRCA2, BTD, CACNA1S, CALM1, CALM2, CALM3, CASQ2, COL3A1, DES, DSC2, DSG2, DSP, ENG, FBN1, FLNC, GAA, GLA, HFE, HNF1A, KCNH2, KCNQ1, LDLR, LMNA, MAX, MEN1, MLH1, MSH2, MSH6, MUTYH, MYBPC3, MYH11, MYH7, MYL2, MYL3, NF2, OTC, PALB2, PCSK9, PKP2, PMS2, PRKAG2, PTEN, RB1, RBM20, RET, RPE65, RYR1, RYR2, SCN5A, SDHAF2, SDHB, SDHC, SDHD, SMAD3, SMAD4, STK11, TGFBR1, TGFBR2, TMEM127, TMEM43, TNNC1, TNNI3, TNNT2, TP53, TPM1, TRDN, TSC1, TSC2, TTN, TTR, VHL, and WT1. Variants unrelated to the patient's phenotype but potentially clinically significant in genes with no known disease association will be reported if elected.

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LIMITATIONS

The interpretation of nucleotide changes is based on our current understanding of the specific genes. This interpretation may change over time as more information about these genes becomes available. Possible diagnostic errors include sample mix-ups, genetic variants that interfere with analysis, incorrect assignment of biological parentage, or other sources. Please contact a genetic counselor at Baylor Genetics if there is reason to suspect one of these sources of error.

REFERENCES

1. Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehm HL; ACMG Laboratory Quality Assurance Committee. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet Med. 2015 May;17(5):405-24. doi: 10.1038/gim.2015.30. Epub 2015 Mar 5. PMID: 25741868; PMCID: PMC4544753.
2. Miller DT, Lee K, Abul-Husn NS, Amendola LM, Brothers K, Chung WK, Gollob MH, Gordon AS, Harrison SM, Hershberger RE, Klein TE, Richards CS, Stewart DR, Martin CL; ACMG Secondary Findings Working Group. ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG). Genet Med. 2023 Aug;25(8):100866. doi: 10.1016/j.gim.2023.100866. PMID: 37347242.

DISCLAIMER

This test was developed, and its performance was determined by the Baylor Miraca Genetics Laboratory DBA Baylor Genetics. It has not been cleared or approved by the U.S. Food and Drug Administration (FDA). Since FDA is not required for clinical use of this test, this laboratory has established and validated the test's accuracy and precision, pursuant to the requirement of CLIA'88. This laboratory is licensed and/or accredited under CLIA and CAP (CAP#2109314 / CLIA# 45D0660090; Lab Director: Christine M. Eng, MD).

ELECTRONICALLY SIGNED BY

Christine M. Eng, M.D. Medical Director	Casey Thornton, Ph.D. Assistant Laboratory Director

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